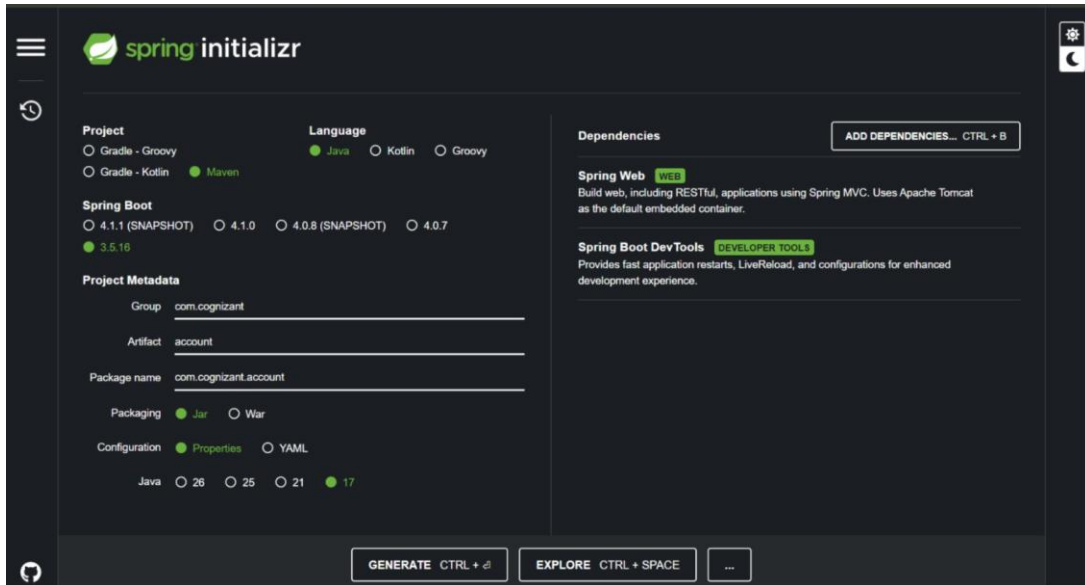


2. Microservices with API gateway

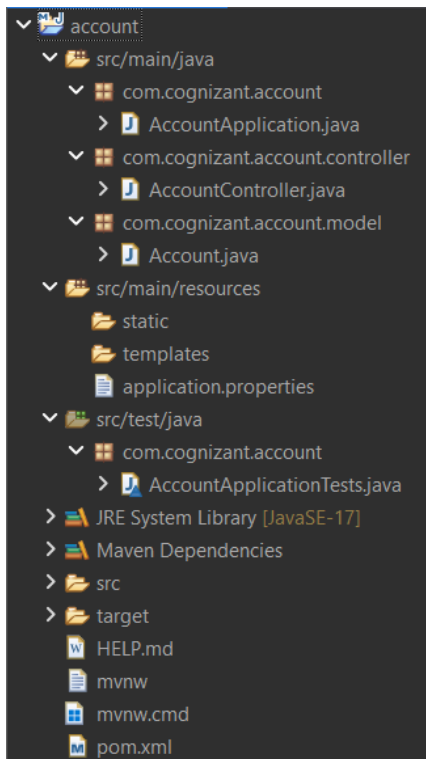
Creating Microservices for Account and Loan

Account Microservice

- Project Details



Project Explorer



Account.java

```
package com.cognizant.account.model;

public class Account {

    private String number;
    private String type;
    private double balance;

    public Account() {
    }

    public Account(String number, String type, double balance) {
        this.number = number;
        this.type = type;
        this.balance = balance;
    }

    public String getNumber() { return number; }
    public void setNumber(String number) { this.number = number; }

    public String getType() { return type; }
    public void setType(String type) { this.type = type; }

    public double getBalance() { return balance; }
    public void setBalance(double balance) { this.balance = balance; }
}
```

AccountController.java

```
package com.cognizant.account.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.account.model.Account;

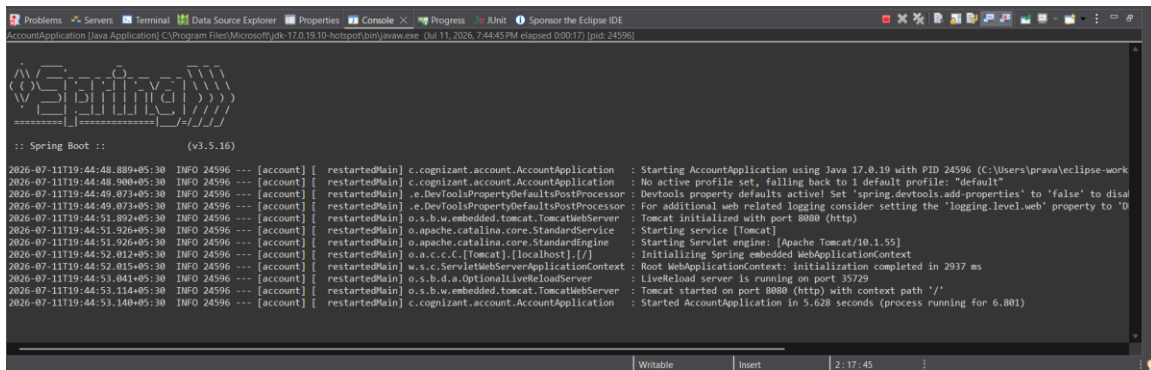
@RestController
public class AccountController {

    @GetMapping("/accounts/{number}")
    public Account getAccount(@PathVariable String number) {
        return new Account(number, "savings", 234343);
    }
}
```

application.properties

```
spring.application.name=account
```

Console Output



```
AccountApplication [Java Application] C:\Program Files\Microsoft\jdk-17.0.19-hotspot\bin\java.exe (Jul 11, 2026, 7:44:45 PM elapsed 0:00:17) [pid: 24596]

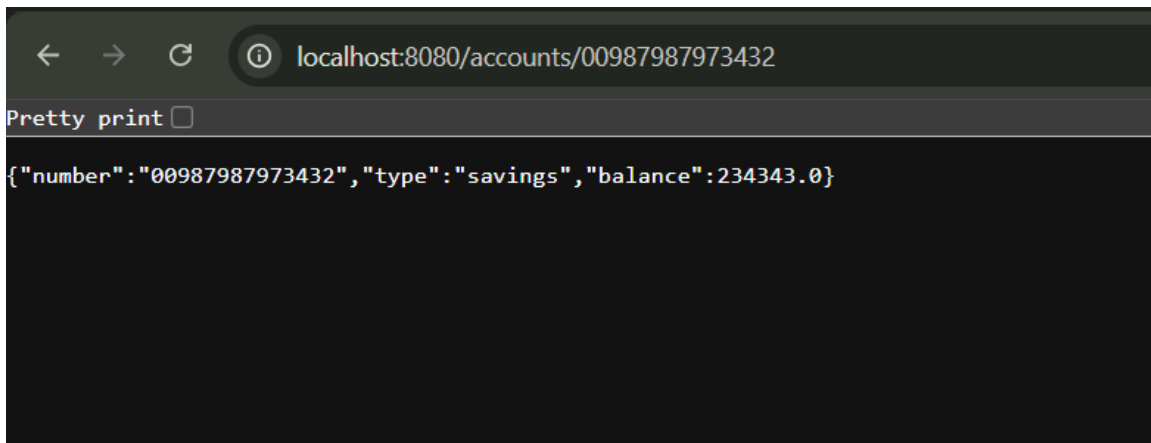
:: Spring Boot :: (v3.5.16)

2026-07-11T19:44:48.889+05:30 INFO 24596 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 17.0.19 with PID 24596 (C:\Users\prava\workspace\AccountApplication\AccountApplication.jar)
2026-07-11T19:44:48.900+05:30 INFO 24596 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-11T19:44:49.073+05:30 INFO 24596 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-11T19:44:49.092+05:30 INFO 24596 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-11T19:44:51.926+05:30 INFO 24596 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-11T19:44:51.926+05:30 INFO 24596 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T19:44:51.926+05:30 INFO 24596 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
2026-07-11T19:44:52.012+05:30 INFO 24596 --- [account] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-11T19:44:52.015+05:30 INFO 24596 --- [account] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2937 ms
2026-07-11T19:44:53.041+05:30 INFO 24596 --- [account] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-11T19:44:53.114+05:30 INFO 24596 --- [account] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-11T19:44:53.140+05:30 INFO 24596 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 5.628 seconds (process running for 6.801)
```

Account Output

URL: <http://localhost:8080/accounts/00987987973432>

```
{"number":"00987987973432","type":"savings","balance":234343.0}
```



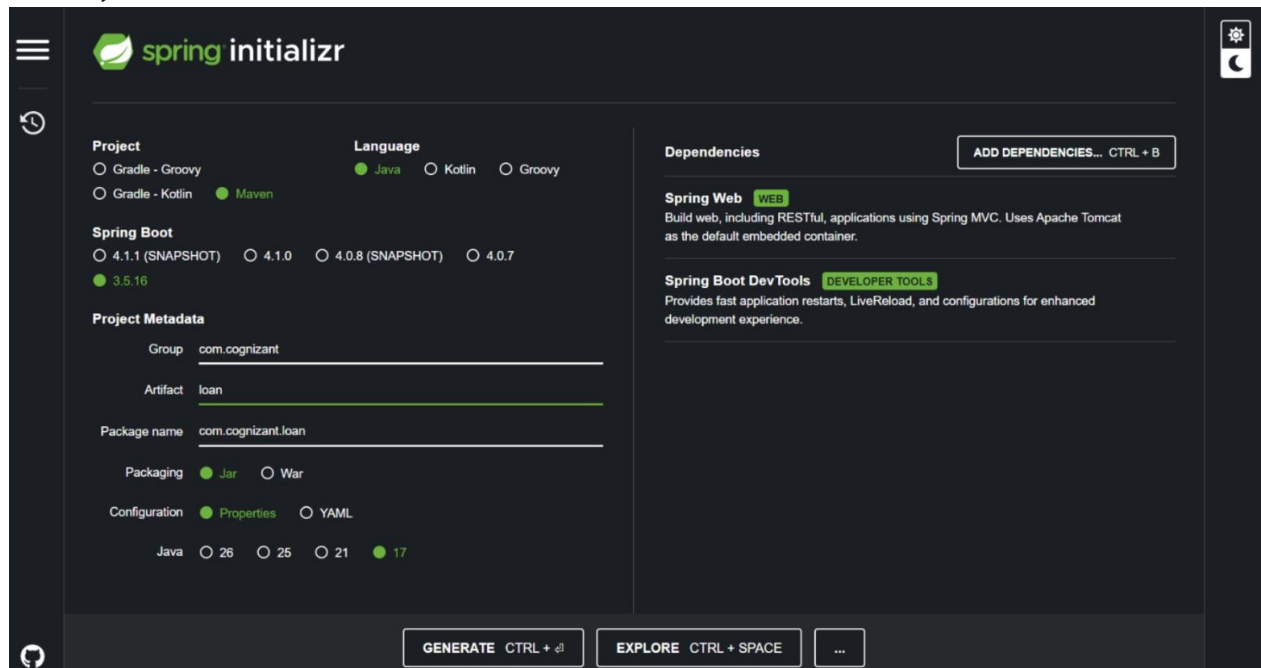
```
localhost:8080/accounts/00987987973432

Pretty print ☐

{"number":"00987987973432","type":"savings","balance":234343.0}
```

Loan Microservice

- Project Details



The screenshot shows the Spring Initializr web interface for configuring a new project. The interface is dark-themed with green accents. On the left, there are navigation icons for a menu, a clock, and GitHub. The main content area is divided into sections for Project, Language, Spring Boot, Project Metadata, and Dependencies. The Project section shows radio buttons for Gradle - Groovy, Gradle - Kotlin, and Maven (selected). The Language section shows radio buttons for Java (selected), Kotlin, and Groovy. The Spring Boot section shows radio buttons for 4.1.1 (SNAPSHOT), 4.1.0, 4.0.8 (SNAPSHOT), 4.0.7, and 3.5.16 (selected). The Project Metadata section has input fields for Group (com.cognizant), Artifact (loan), and Package name (com.cognizant.loan). The Packaging section has radio buttons for Jar (selected) and War. The Configuration section has radio buttons for Properties (selected) and YAML. The Java section has radio buttons for 26, 25, 21, and 17 (selected). The Dependencies section has a button to ADD DEPENDENCIES... (CTRL + B) and lists two dependencies: Spring Web (WEB) and Spring Boot DevTools (DEVELOPER TOOLS). At the bottom, there are buttons for GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), and a menu icon.

Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ Maven

Language

☒ Java ☐ Kotlin ☐ Groovy

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☐ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☒ 3.5.16

Project Metadata

Group

Artifact

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

Java ☐ 26 ☐ 25 ☐ 21 ☒ 17

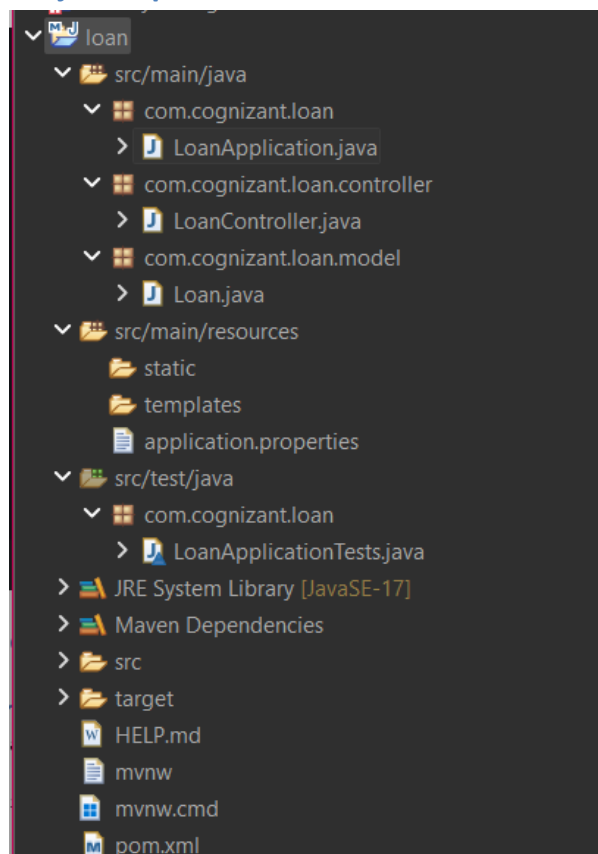
Dependencies [ADD DEPENDENCIES... CTRL + B](#)

Spring Web [WEB](#)
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools [DEVELOPER TOOLS](#)
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

[GENERATE CTRL + G](#) [EXPLORE CTRL + SPACE](#) [...](#)

Project Explorer



Loan.java

```
package com.cognizant.loan.model;

public class Loan {

    private String number;
    private String type;
    private double loan;
    private double emi;
    private int tenure;

    public Loan() {
    }

    public Loan(String number, String type, double loan, double emi, int
tenure) {
        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }

    public String getNumber() { return number; }
    public void setNumber(String number) { this.number = number; }

    public String getType() { return type; }
    public void setType(String type) { this.type = type; }

    public double getLoan() { return loan; }
    public void setLoan(double loan) { this.loan = loan; }

    public double getEmi() { return emi; }
    public void setEmi(double emi) { this.emi = emi; }

    public int getTenure() { return tenure; }
    public void setTenure(int tenure) { this.tenure = tenure; }
}
```

LoanController.java

```
package com.cognizant.loan.controller;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.loan.model.Loan;

@RestController
public class LoanController {

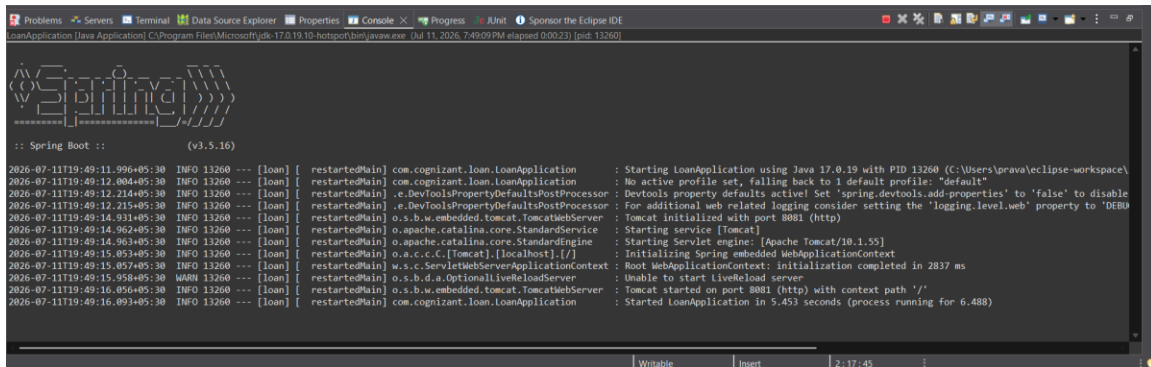
    @GetMapping("/loans/{number}")
    public Loan getLoan(@PathVariable String number) {
```

```

    return new Loan(number, "car", 400000, 3258, 18);
}
}

```

Console Output



```

LoanApplication [Java Application] C:\Program Files\Microsoft\jdk-17.0.19-hotspot\bin\java.exe (Jul 11, 2026, 7:49:09 PM elapsed 0:00:23) [pid: 13260]

:: Spring Boot :: (v3.5.16)

2026-07-11T19:49:11.996+05:30 INFO 13260 --- [loan] [ restarted@Main] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 17.0.19 with PID 13260 (C:\Users\prava\workspace\
2026-07-11T19:49:12.004+05:30 INFO 13260 --- [loan] [ restarted@Main] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-11T19:49:12.214+05:30 INFO 13260 --- [loan] [ restarted@Main] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-11T19:49:12.215+05:30 INFO 13260 --- [loan] [ restarted@Main] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-11T19:49:14.931+05:30 INFO 13260 --- [loan] [ restarted@Main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-11T19:49:14.962+05:30 INFO 13260 --- [loan] [ restarted@Main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T19:49:14.963+05:30 INFO 13260 --- [loan] [ restarted@Main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
2026-07-11T19:49:15.053+05:30 INFO 13260 --- [loan] [ restarted@Main] o.a.c.c.C.[Tomcat].[localhost] [/] : Initializing Spring embedded WebApplicationContext
2026-07-11T19:49:15.057+05:30 INFO 13260 --- [loan] [ restarted@Main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2837 ms
2026-07-11T19:49:15.958+05:30 WARN 13260 --- [loan] [ restarted@Main] o.s.b.d.a.OptionalLiveReloadServer : Unable to start LiveReload server
2026-07-11T19:49:16.056+05:30 INFO 13260 --- [loan] [ restarted@Main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-11T19:49:16.093+05:30 INFO 13260 --- [loan] [ restarted@Main] com.cognizant.loan.LoanApplication : Started LoanApplication in 5.453 seconds (process running for 6.488)

```

application.properties

```

spring.application.name=loan
server.port=8081

```

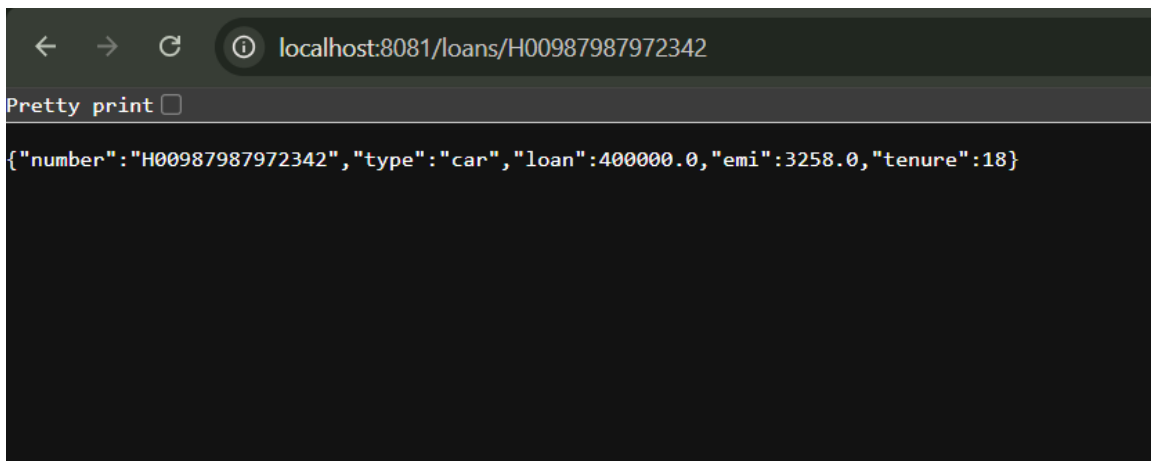
Loan Output

URL: <http://localhost:8081/loans/H00987987972342>

```

{"number":"H00987987972342","type":"car","loan":400000.0,"emi":3258.0,"tenure":18}

```



```

localhost:8081/loans/H00987987972342

Pretty print

{"number":"H00987987972342","type":"car","loan":400000.0,"emi":3258.0,"tenure":18}

```